

Is the environmental impact of AI overstated?

Posted by u/valonianfool • 0 points • 29 comments

I've found conflicting sources and arguments about the environmental impact of AI. On one hand, proponents argue that in comparison to other industries such as agriculture AI is a drop in the bucket when it comes to water and energy consumed, on the other hand I've read sources saying that training data centers in Texas are consuming a great deal of water amid a drought while people are asked to cut down on their personal water use: [Texas AI centers guzzle 463 million gallons, now residents are asked to cut back on showers] (https://m.economictimes.com/news/international/us/texas-ai-data-centers-water-usage-texas-ai-centers-guzzle-463-million-gallons-now-residents-are-asked-to-cut-back-on-showers-ai-news/amp_article/show/122983253.cms)

Though I don't understand the statement from the article that water that is evaporated "is lost forever", because unless it's stored in an enclosed space like a tank all water is part of a cycle.

Furthermore, "Artificial intelligence tools trained with large language models, such as Open AI's ChatGPT, among others, use exponentially more computing power than traditional cloud services.": [Data Drain: The Land and Water Impact of the AI Boom](<https://www.lincolnst.edu/publications/land-lines-magazine/articles/land-water-impacts-data-centers/>)

****Is the environmental impact of generative ai overstated because it's still small in comparison to other industries?****

Comments

u/Ok-Pea-4838 • 1 points • 2026-01-21 14:42 UTC

A lot of it is overstated because training, location, and cooling methods get mixed together. Comparing cited sources side by side usually shows the scale more clearly. I sometimes use AlchemistAiOfficial for that since it surfaces citations and efficiency context instead of single scary numbers.

u/btxcfms • 1 points • 2026-01-20 10:32 UTC

It is severely understated because ppl tend to consider different metrics and not the cumulative effect. It is long past being simply problematic, but for now let the mass think that this is just "debatable". Actually you asked in a right time. 2026 will be the year of noticing something is not right here.

u/Apart-Kangaroo-7648 • 1 points • 2026-01-20 02:18 UTC

Overstated? Yes.

Problematic? Yes.

Dwarfed by the worst offending industries? Yes.

u/Chaghatai • 0 points • 2026-01-17 13:02 UTC

Yes, massively so

When it comes to actual environmental damage from depriving raisins of water that does not immediately go back into the local water cycle. Look no further than agriculture

People growing crops in deserts cause rivers to go down levels far lower than natural

When it comes to a comparison with AI water use, it's not even close

People who are complaining about AI water use but are not complaining about almond plantations, mega golf courses, and the American lawn are being performative

u/KeldTundraking • 1 points • 2026-01-17 12:26 UTC

I'm not sure the waifu slop generator should be compared to agriculture.

We die without food.

u/BarberCompetitive517 • 1 points • 2026-01-16 18:17 UTC

No, it is not overstated. It is a severe problem that AI boosters want to sweep under the rug because all they are is gamblers trying to make money on the stock.

u/Vaughn • 2 points • 2026-01-16 13:33 UTC

The impact isn't zero, but you need to be careful not to prove too much. Things with more environmental impact per (for the sake of argument) image:

- \- Eating a single hamburger (by a factor of 1000).
- \- Making the paper used to draw a similar image. A single A4 sheet, specifically.

That doesn't make environmental concerns go away, but environmental concerns should be considered in proportion to the benefit, and the fundamental benefit of everything we do is essentially "it makes someone happy". ...unfortunately sometimes the only person who's made happy is a billionaire, but that's not the case for e.g. someone generating an image of their OC at home.

u/valonianfool • 1 points • 2026-01-16 13:52 UTC

Evidence that eating a single hamburger has a higher impact? Because the energy, water and emissions attributed to beef have been greatly exaggerated. Most water attributed to livestock production is green water, or rainwater that falls on a field, and soy isn't grown specifically for livestock, the soy in animal feed is soy meal which is what's left over after producing oil.

u/Ok_Dog_7189 • 1 points • 2026-01-18 11:06 UTC

Jumping in as the "trust me bro" reddit expert... But the big problem with mass livestock production are

- Deforestation. Grass monocultures are far worse than you think for biodiversity... I mean fields are an ecological disaster on an incomprehensible scale since we're so accustomed to them... Plus replacing trees and bushes with fields don't hold water, they're awful for causing flooding.
- Eutrophication... Same as flooding, cow shit basically runs straight into water supplies, builds up nutrients and in the wrong weather conditions causes algal blooms and wipes out ecosystems... Maybe a drought where the water isn't running fast... Or hits a dam... Pretty quickly wipes out an ecosystem where not managed correctly.
- Energy. Freezing and cooling meat for storage and transport is where the energy goes... I can't tell you the exact figure, but we're looking on par to the equivalent of a country's energy supply just for that.

You don't have to be against meat, but I would personally suggest to aim to get locally sourced meat where feasible to mitigate this a little.

u/valonianfool • 1 points • 2026-01-18 12:28 UTC

The vast majority of deforestation occurs in Tropical countries like sea and south America, and tropical Rainforests are the wrong environment for cattle anyway. It has nothing to do with Europe or north america. Lands grazed by Cattle can become carbon sinks and livestock and wild animal species can coexist in a biodiverse ecosystem.

Crop monocultures are just as bad.

u/Ok_Dog_7189 • 1 points • 2026-01-18 13:15 UTC

are you joking? :\

u/valonianfool • 1 points • 2026-01-18 13:36 UTC

Can you tell me what about my statement is wrong instead of this?

u/Ok_Dog_7189 • 1 points • 2026-01-18 16:51 UTC

Everything... Deforestation for cattle is the biggest reason the Amazon is being burned... It expanded only a few years ago under Bolsonaro... Cows don't live in the jungle so they burn it down for them.

US, Europe Asia already went though deforestation for agriculture and industry, Europe was once an almost entirely forested continent... We already deforested it that's why it doesn't happen here anymore.

Grassland is technically a carbon sink, but not great at it compared to dense vegetation and wildlife in general is not compatible with livestock although some thrive and some have adapted it's not a natural environment.

And yes growing crops is also bad... But essentially takes up significantly less space per yield and doesn't require the same level of freezing to preserve and transport, so is far more sustainable whilst still being bad

u/valonianfool

• 1
points
•
2026-01-18
17:43
UTC

I
never
denied
that
the
Amazon
Rainforest
is
being
deforested
for
cattle
but
your

statement about north america and Europe are moot points.

Source for "grasslands aren't great for carbon sinks compared to sense vegetation"? Because Natural grasslands do exist and livestock production can be done sustainably.

Even in south America, in the pantanal region cattle ranching occurs in a very biodiverse area alongside many endangered species.

u/Isopod_Danger_42069 • 3 points • 2026-01-16 07:14 UTC

It's real but it's also exaggerated. Those aren't "ai data centers" they're just data centers. Ai related activities only account for a small percentage of their operations, different estimates I've seen ranged from 13 to 20%.

Most of is everyday webuse, with gaming, streaming, and torrenting being much larger components of their overall activities. The

truth is that we're a data hungry society and ai is, to a degree, being scapegoated.

Let me put it this way, for all the people here worried about the environmental cost of ai, would you be willing to give up your streaming services, gaming, social media (yes, social media is also bad for the environment), and torrenting to help the environment? Because that would actually make a much bigger difference, or are you only concerned about ai because getting rid of it wouldn't require you personally to give anything up?